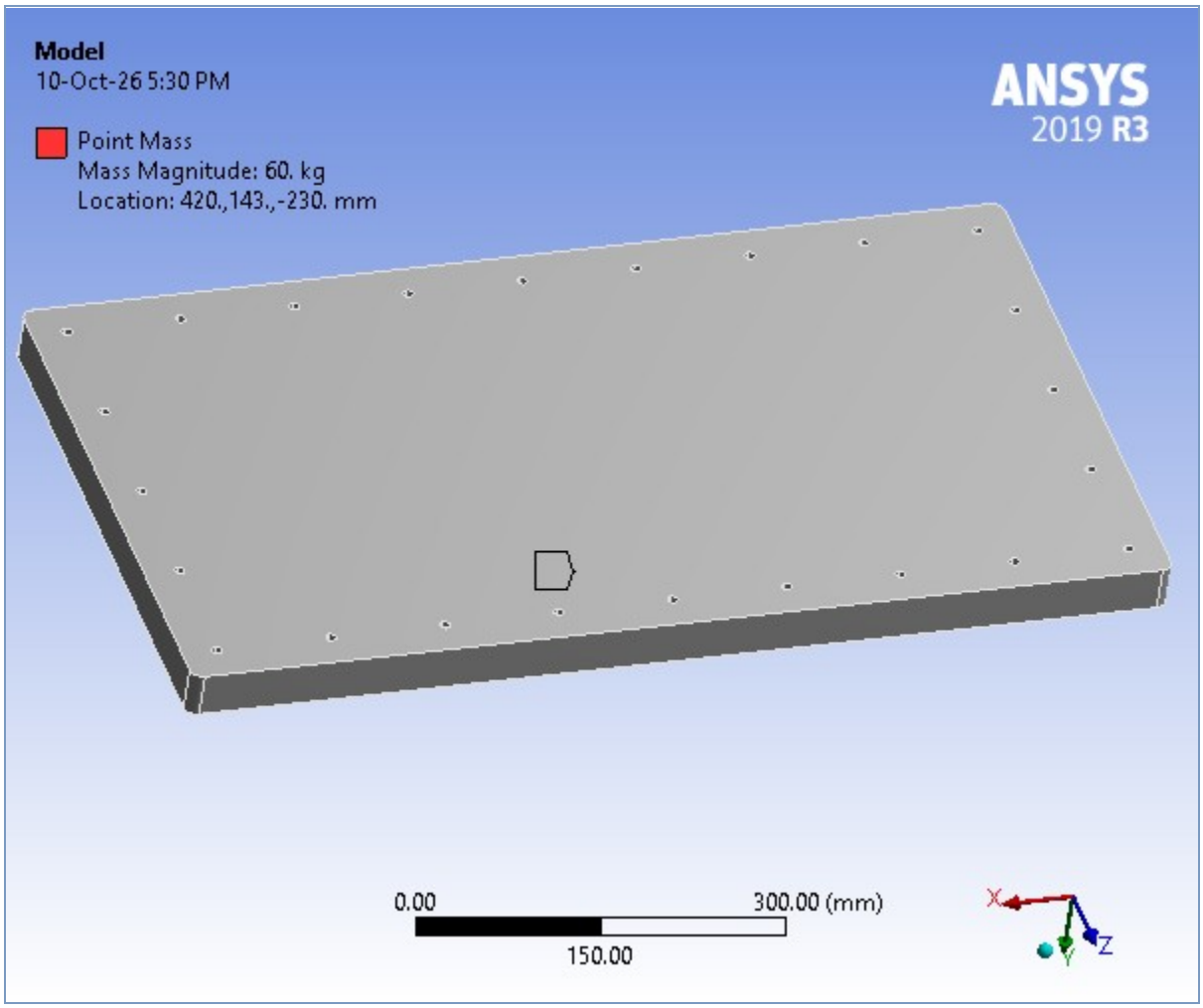
Project

First Saved	Saturday, October 10, 2026
Last Saved	Saturday, October 10, 2026
Product Version	2019 R3
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (A4)**
  - Geometry
    - battery_test_fixture_plate-FreeParts|Interface Holes M10
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  - Coordinate Systems
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  - **Modal (A5)**
    - Pre-Stress (None)
    - Analysis Settings
    - Fixed Support
    - Solution (A6)
      - Solution Information
      - Total Deformation
- **Material Data**
  - Al 6061-T6

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

Geometry

TABLE 2
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\kaila\Downloads\portfolio work\battery_test_fixture_plate.STEP
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	840. mm
Length Y	40. mm
Length Z	460. mm
Properties	
Volume	1.5414e+007 mm³

Mass	101.62 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	126008
Elements	74170
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 3
Model (A4) > Geometry > Parts

Object Name	*battery_test_fixture_plate-FreeParts|Interface Holes M10*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Al 6061-T6
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	840. mm

Length Y	40. mm
Length Z	460. mm
Properties	
Volume	1.5414e+007 mm³
Mass	41.619 kg
Centroid X	420. mm
Centroid Y	20.013 mm
Centroid Z	-230. mm
Moment of Inertia Ip1	7.38e+005 kg·mm²
Moment of Inertia Ip2	3.1762e+006 kg·mm²
Moment of Inertia Ip3	2.4493e+006 kg·mm²
Statistics	
Nodes	126008
Elements	74170
Mesh Metric	None

TABLE 4
Model (A4) > Geometry > Masses

Object Name	*Point Mass*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Applied By	Remote Attachment
Geometry	12 Faces
Coordinate System	Global Coordinate System
X Coordinate	420. mm
Y Coordinate	143. mm
Z Coordinate	-230. mm
Location	Defined
Definition	
Mass	60. kg
Mass Moment of Inertia X	0. kg·mm²
Mass Moment of Inertia Y	0. kg·mm²
Mass Moment of Inertia Z	0. kg·mm²
Suppressed	No
Behavior	Deformable
Pinball Region	All

TABLE 5
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 6
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined

Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Mesh

TABLE 7
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	6.0 mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	958.54 mm
Average Surface Area	8939.8 mm²
Minimum Edge Length	13.352 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Quality	Default (0.050000)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No

Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	126008
Elements	74170

Modal (A5)

TABLE 8
Model (A4) > Analysis

Object Name	*Modal (A5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Modal
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 9
Model (A4) > Modal (A5) > Initial Condition

Object Name	*Pre-Stress (None)*
State	Fully Defined
Definition	
Pre-Stress Environment	None Available

TABLE 10
Model (A4) > Modal (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Options	
Max Modes to Find	10
Limit Search to Range	Yes
Range Minimum	0. Hz
Range Maximum	2000. Hz
Solver Controls	
Damped	No
Solver Type	Program Controlled
Rotordynamics Controls	
Coriolis Effect	Off
Campbell Diagram	Off
Output Controls	
Stress	No
Surface Stress	No
Back Stress	No
Strain	No
Contact Data	Yes
Nodal Forces	No

Calculate Reactions	No
General Miscellaneous	No
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\kaila\Downloads\portfolio work\fixture_modal_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 11
Model (A4) > Modal (A5) > Loads

Object Name	*Fixed Support*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	48 Faces
Definition	
Type	Fixed Support
Suppressed	No

Solution (A6)

TABLE 12
Model (A4) > Modal (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	1 m 0 s
MAPDL Memory Used	1.8359 GB
MAPDL Result File Size	50.188 MB
Post Processing	
Beam Section Results	No

The following bar chart indicates the frequency at each calculated mode.

FIGURE 1
Model (A4) > Modal (A5) > Solution (A6)

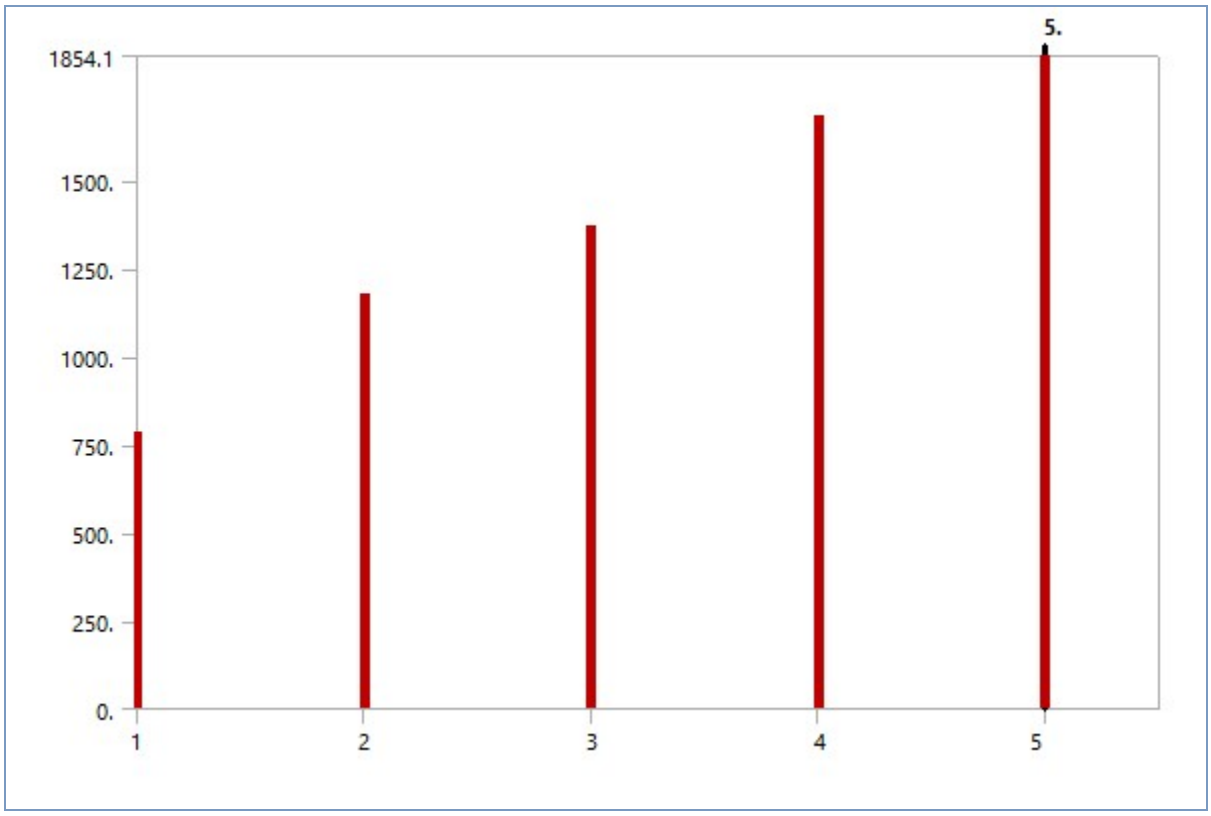


TABLE 13
Model (A4) > Modal (A5) > Solution (A6)

Mode	Frequency [Hz]
1.	785.87
2.	1179.6
3.	1370.4
4.	1681.4
5.	1854.1

TABLE 14
Model (A4) > Modal (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 15
Model (A4) > Modal (A5) > Solution (A6) > Results

Object Name	*Total Deformation*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Total Deformation
Mode	1.
Identifier	
Suppressed	No
Results	
Minimum	0. mm
Maximum	7.9146 mm
Average	2.6013 mm
Minimum Occurs On	battery_test_fixture_plate-FreeParts|Interface Holes M10
Maximum Occurs On	battery_test_fixture_plate-FreeParts|Interface Holes M10
Information	
Frequency	785.87 Hz

TABLE 16
Model (A4) > Modal (A5) > Solution (A6) > Total Deformation

Mode	Frequency [Hz]
1.	785.87
2.	1179.6
3.	1370.4
4.	1681.4
5.	1854.1

Material Data

Al 6061-T6

TABLE 17
Al 6061-T6 > Constants

Density	2.7e-006 kg mm^-3

TABLE 18
Al 6061-T6 > Color

Red	Green	Blue
181	194	156

TABLE 19
Al 6061-T6 > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
68900	0.33	67549	25902	